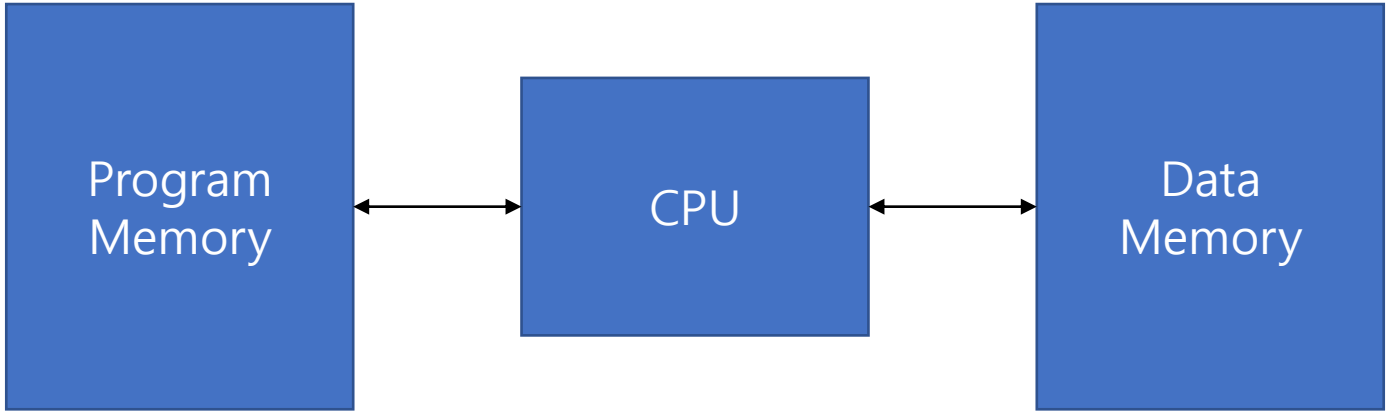
Digital IO

마이크로프로세서 종합 설계. 4주차.

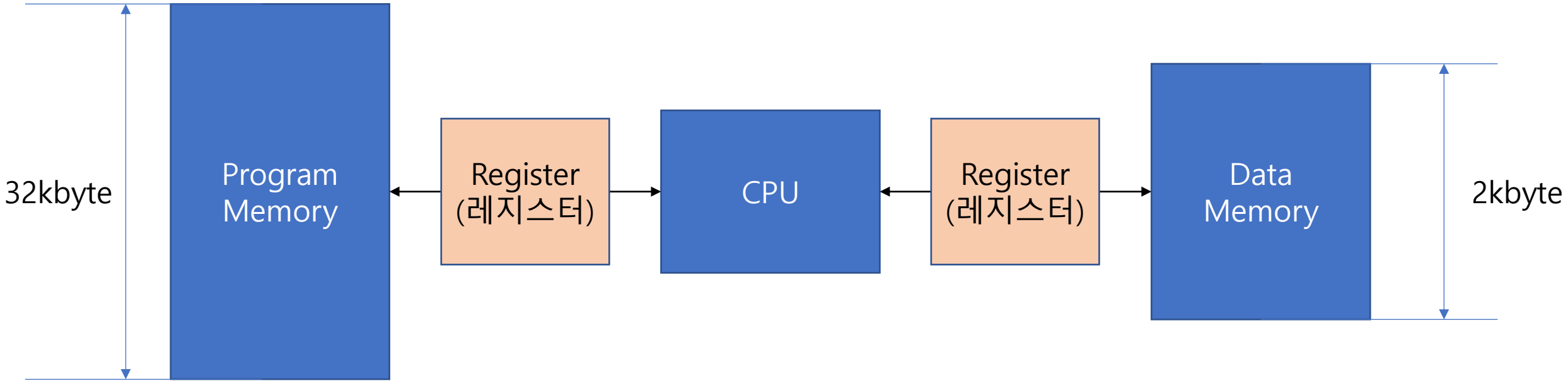
오늘의 목표

1. **디지털 입출력 레지스터의 이해**
2. **디지털 입출력 실험**

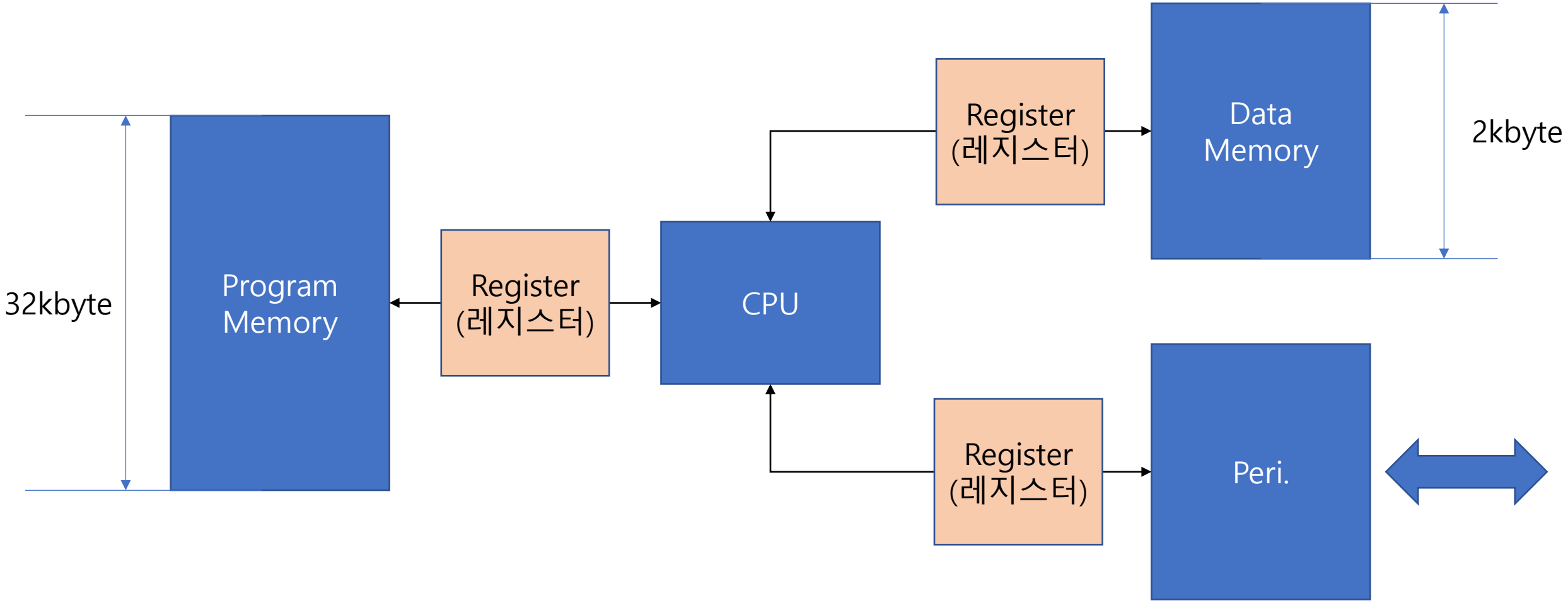
하버드 구조(Harvard architecture)



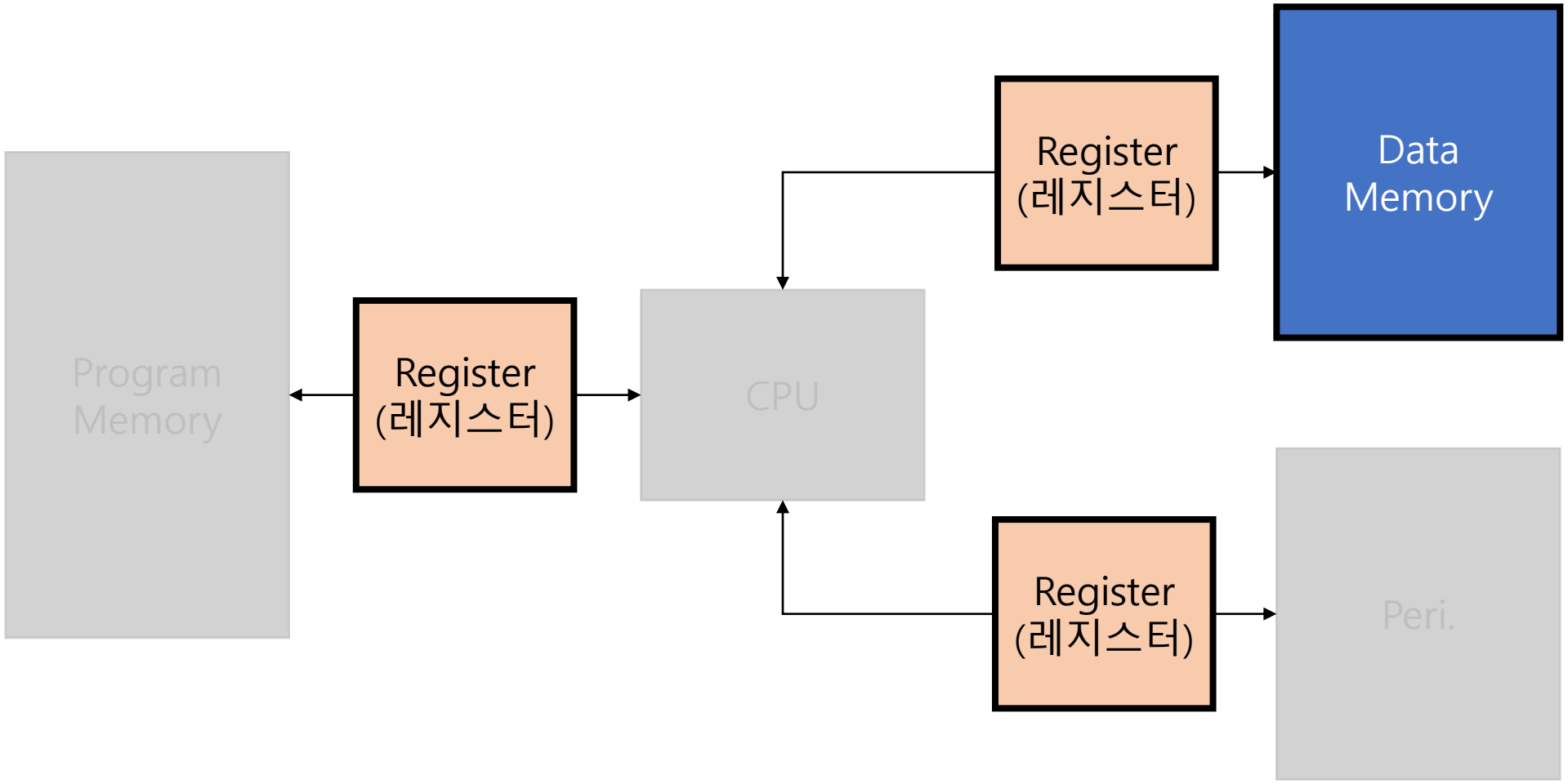
Atmega328p의 메모리



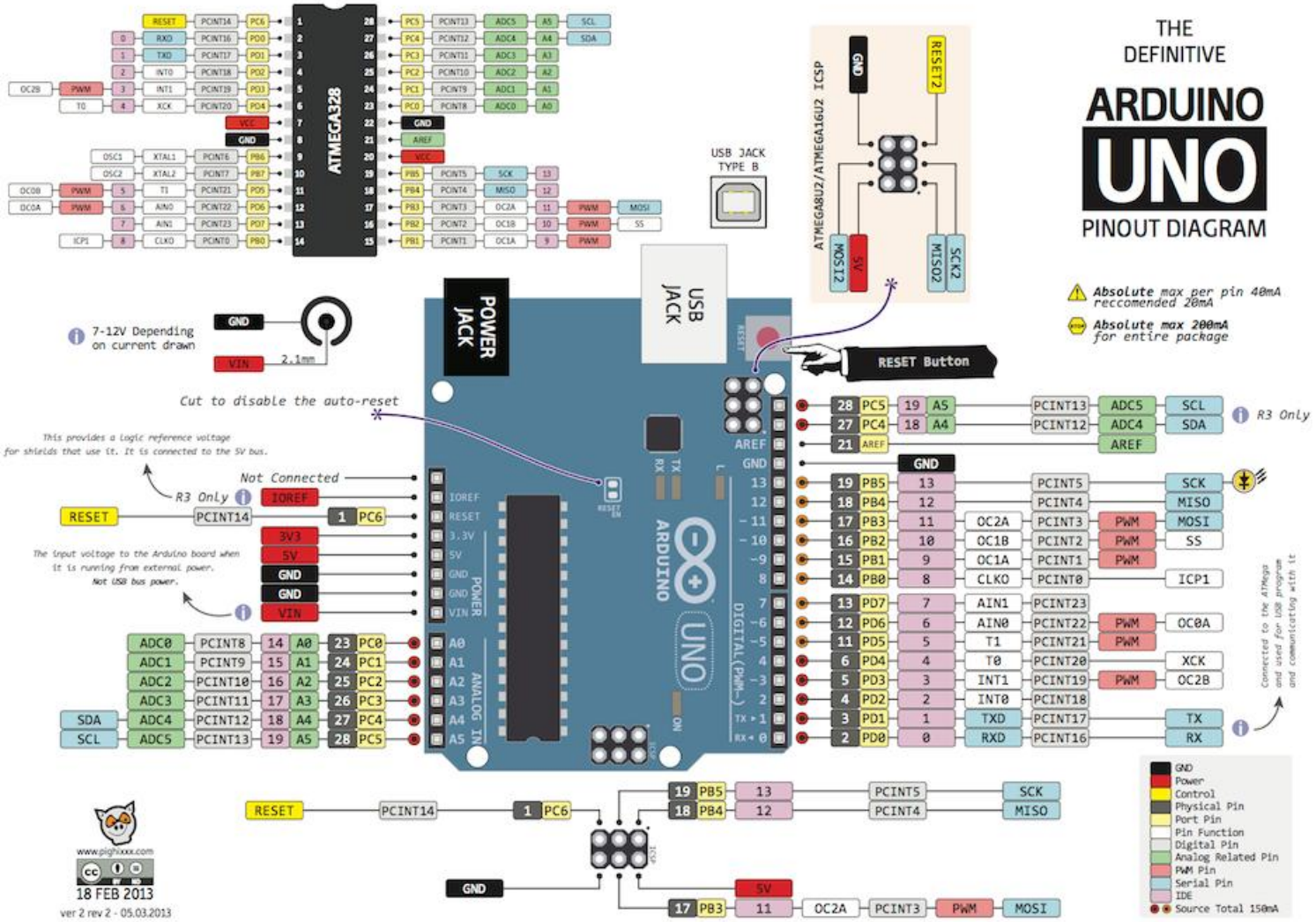
Atmega328p의 메모리 & 외부장치

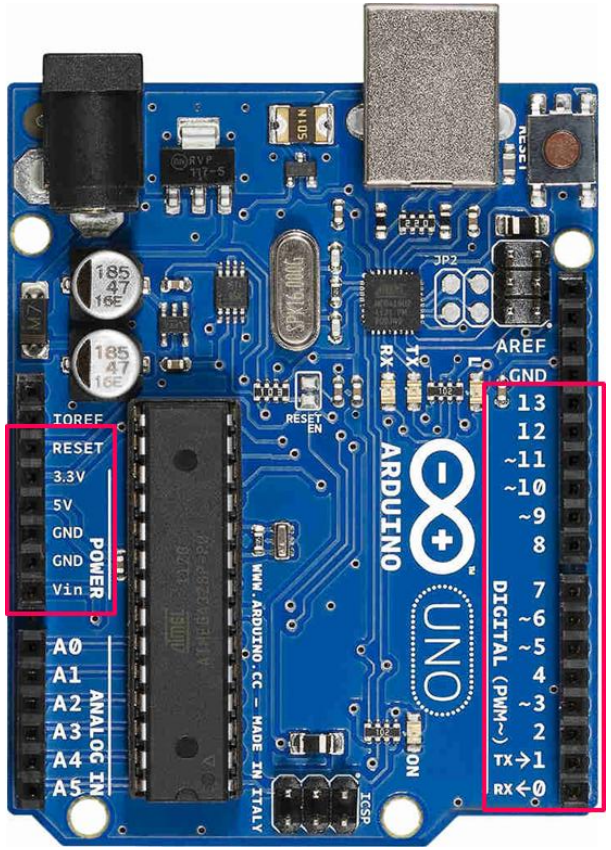


Atmega328p의 메모리맵

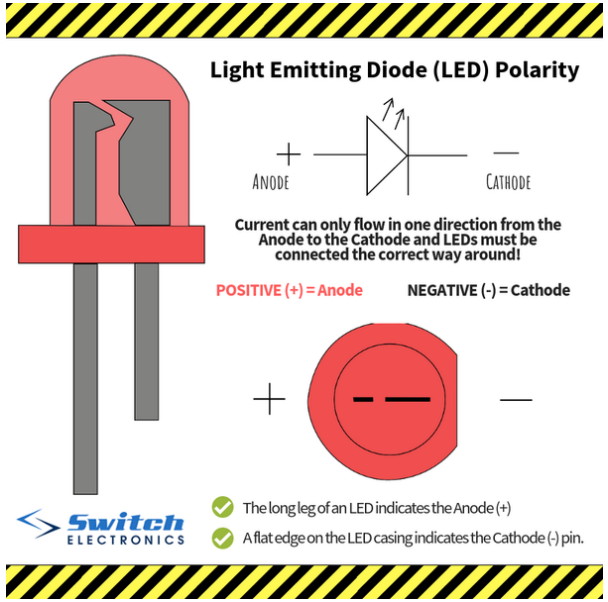


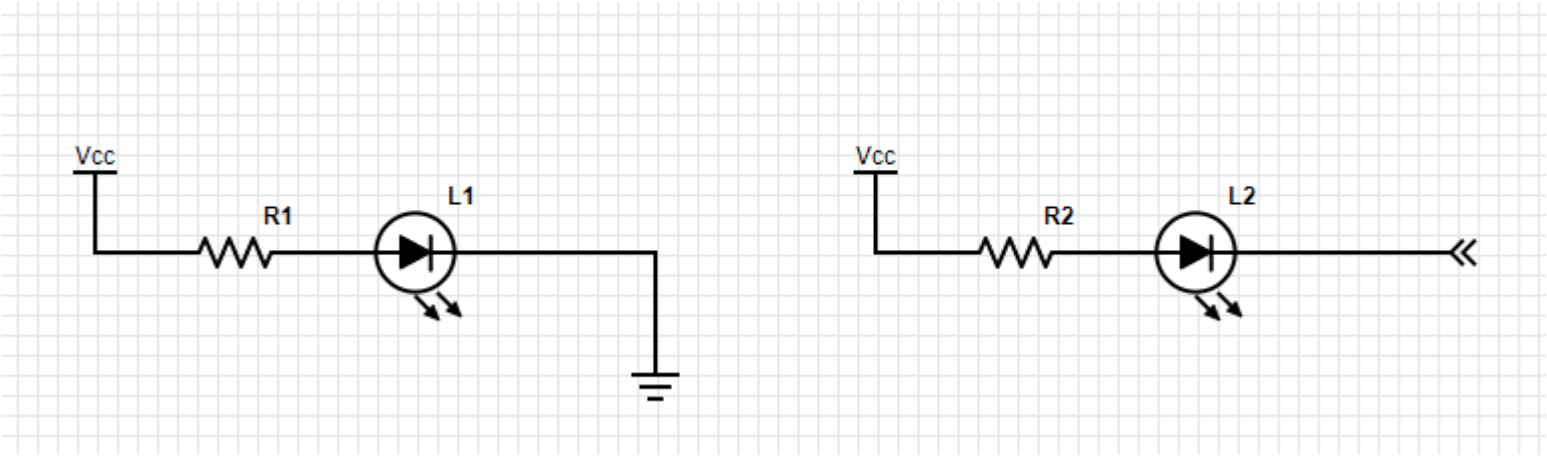
아두이노 디지털 IO


THE DEFINITIVE
ARDUINO
UNO
PINOUT DIAGRAM
ATMEGA328
ATMEGA8U2/ATMEGA16U2 ICSP
USB JACK TYPE B
POWER JACK
USB JACK
RESET Button
7-12V Depending on current drawn
Cut to disable the auto-reset
Not Connected
R3 Only
Absolute max per pin 40mA reccomended 20mA
Absolute max 200mA for entire package
18 FEB 2013
ver 2 rev 2 - 05.03.2013


IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
DIGITAL (PWM~)
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
ARDUINO
UNO
WWW.ARDUINO.CC – MADE IN ITALY

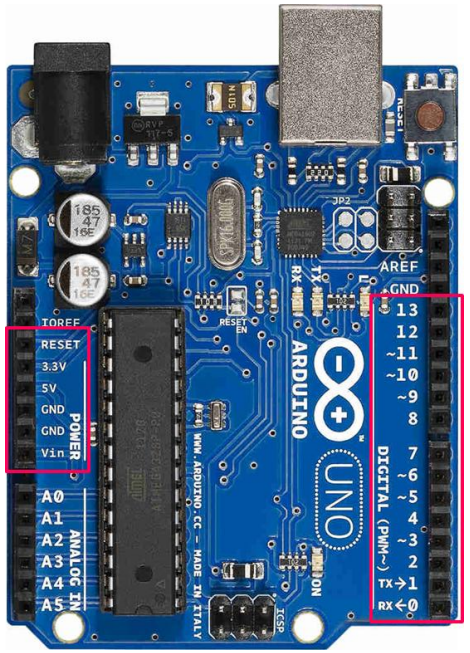
아두이노 디지털 IO

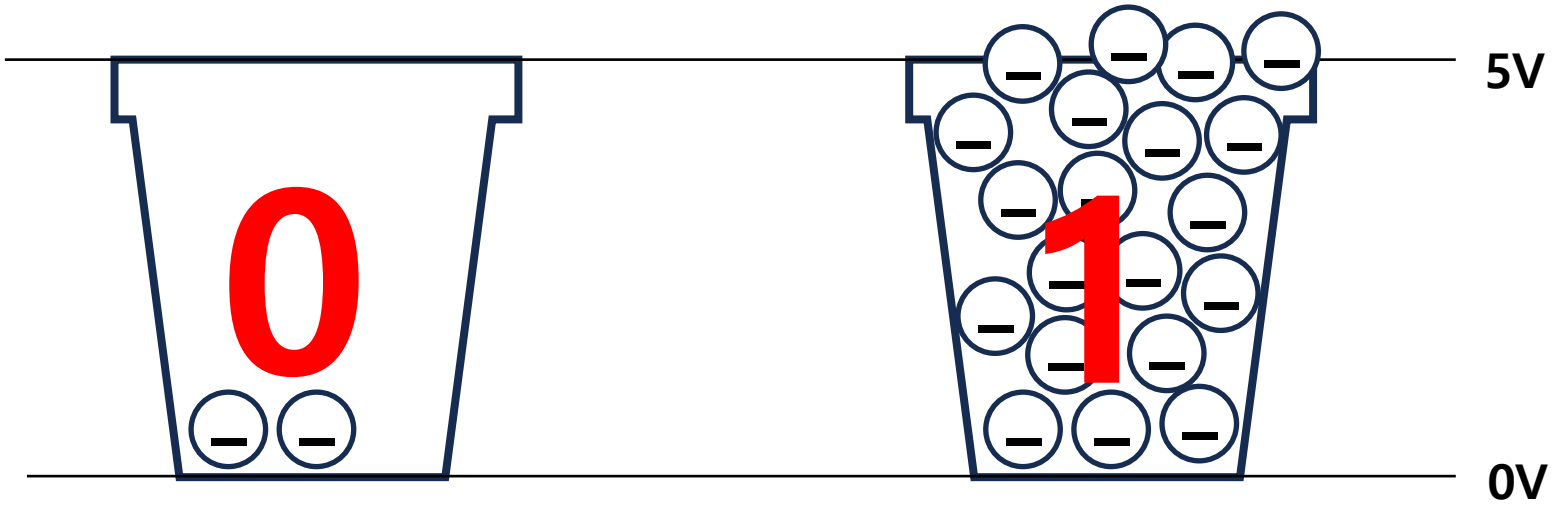

Light Emitting Diode (LED) Polarity
+
ANODE
—
CATHODE
Current can only flow in one direction from the Anode to the Cathode and LEDs must be connected the correct way around!
POSITIVE (+) = Anode
NEGATIVE (-) = Cathode
+
—
Switch ELECTRONICS
The long leg of an LED indicates the Anode (+)
A flat edge on the LED casing indicates the Cathode (-) pin.


Vcc
R1
L1
Vcc
R2
L2

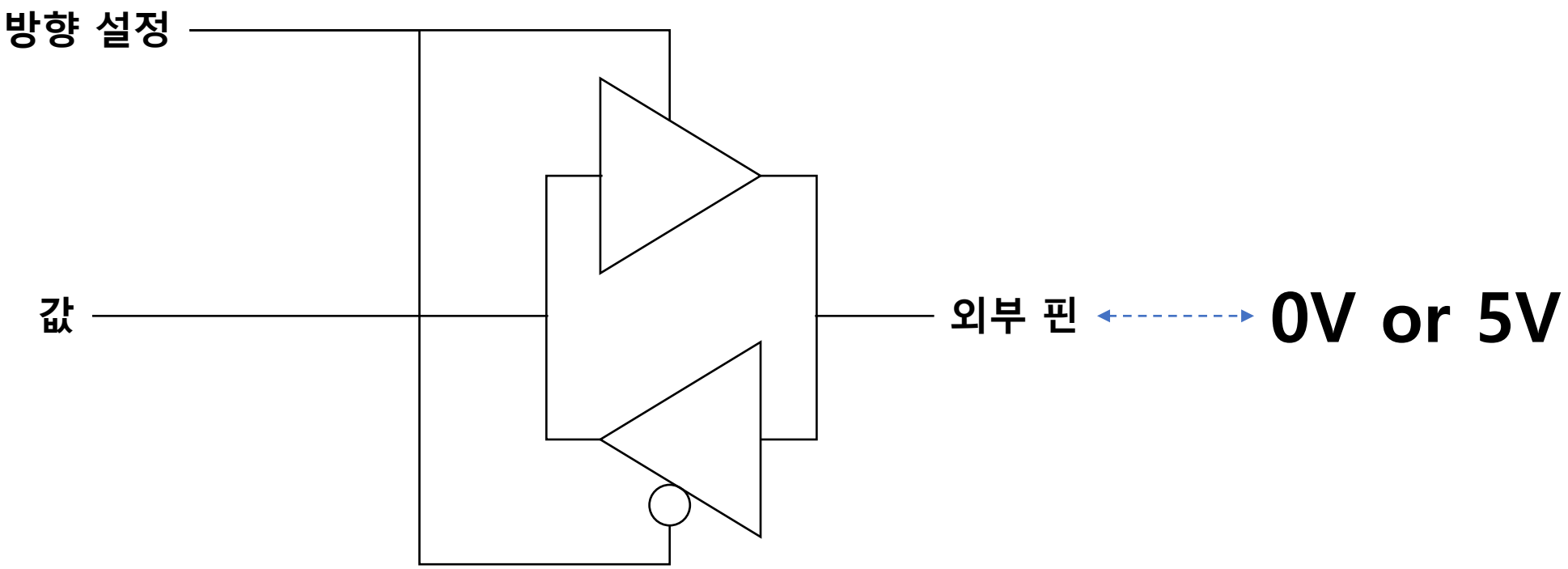
디지털 IO(Input/Output)

- 전자 장치 및 시스템에서 **신호를 주고받는 기본적인 방식**
- 신호는 **0(LOW, 0V)** 또는 **1(HIGH, 5V)**로 표현
- 다양한 센서, 액추에이터, 마이크로컨트롤러, PLC 등에서 사용





디지털 IO(Input/Output)



디지털 IO 포트 관련 레지스터

- Port

13.4.2 PORTB – The Port B Data Register

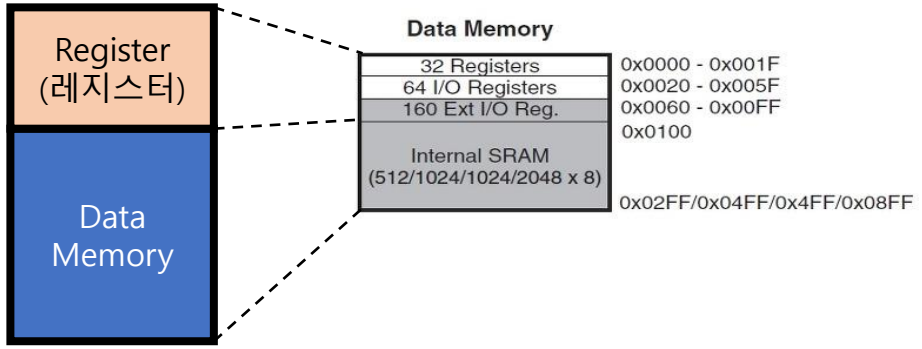
Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

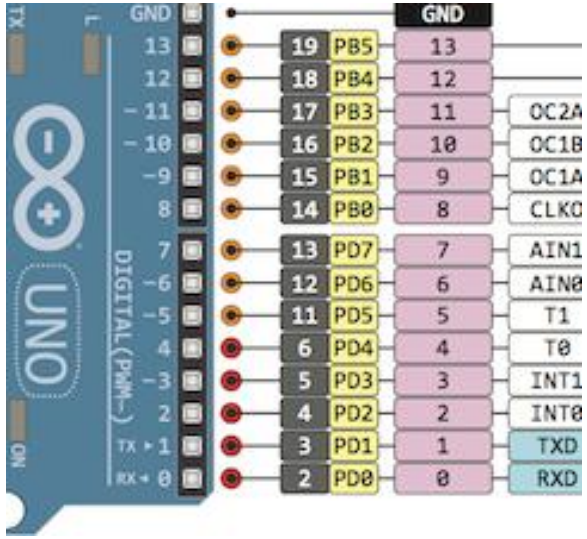
13.4.3 DDRB – The Port B Data Direction Register

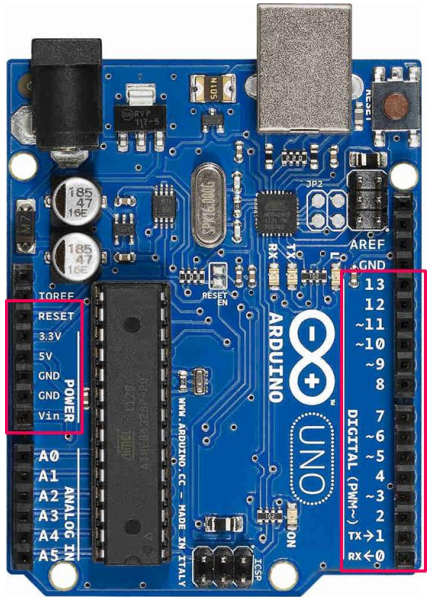
Bit	7	6	5	4	3	2	1	0	
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.4 PINB – The Port B Input Pins Address

Bit	7	6	5	4	3	2	1	0	
0x03 (0x23)	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	PINB
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

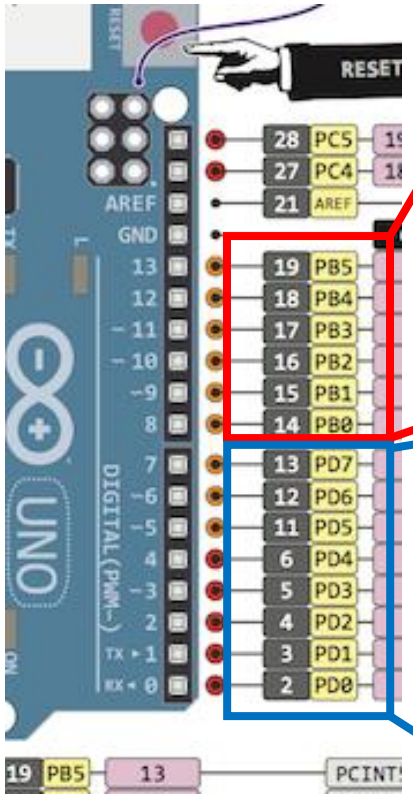






디지털 IO 포트 관련 레지스터

- Port



13.4.2 PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.3 DDRB – The Port B Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.4 PINB – The Port B Input Pins Address

Bit	7	6	5	4	3	2	1	0	
0x03 (0x23)	PINB7	PINB6	PINB5	PINB4	PINB3	PINB2	PINB1	PINB0	PINB
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

13.4.8 PORTD – The Port D Data Register

Bit	7	6	5	4	3	2	1	0	
0x0B (0x2B)	PORTD7	PORTD6	PORTD5	PORTD4	PORTD3	PORTD2	PORTD1	PORTD0	PORTD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.9 DDRD – The Port D Data Direction Register

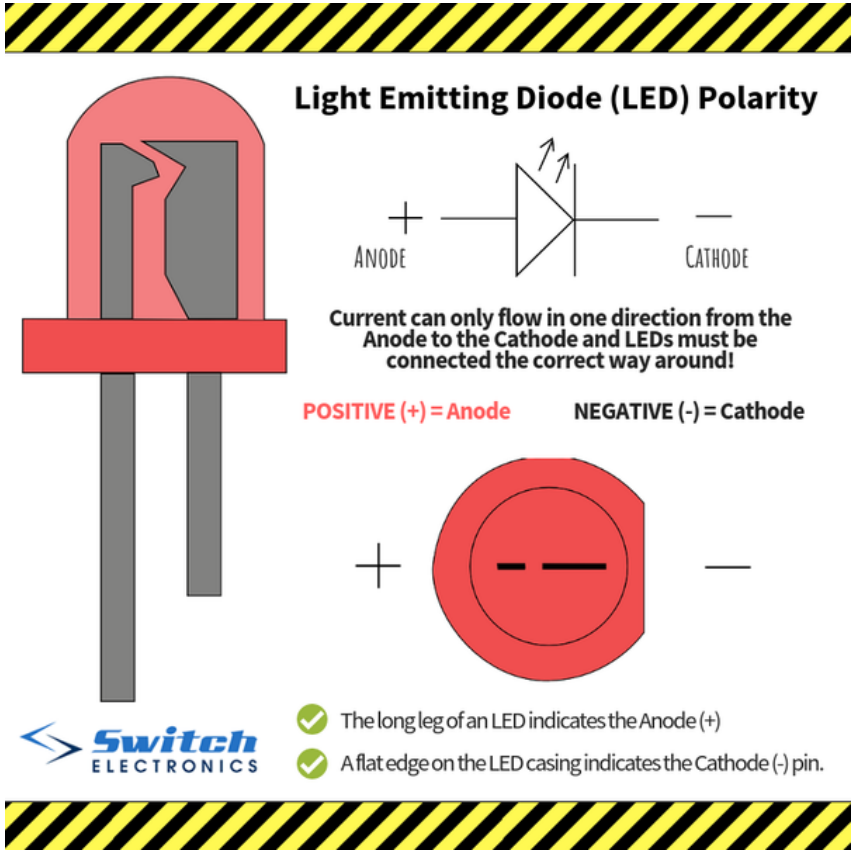
Bit	7	6	5	4	3	2	1	0	
0x0A (0x2A)	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

13.4.10 PIND – The Port D Input Pins Address

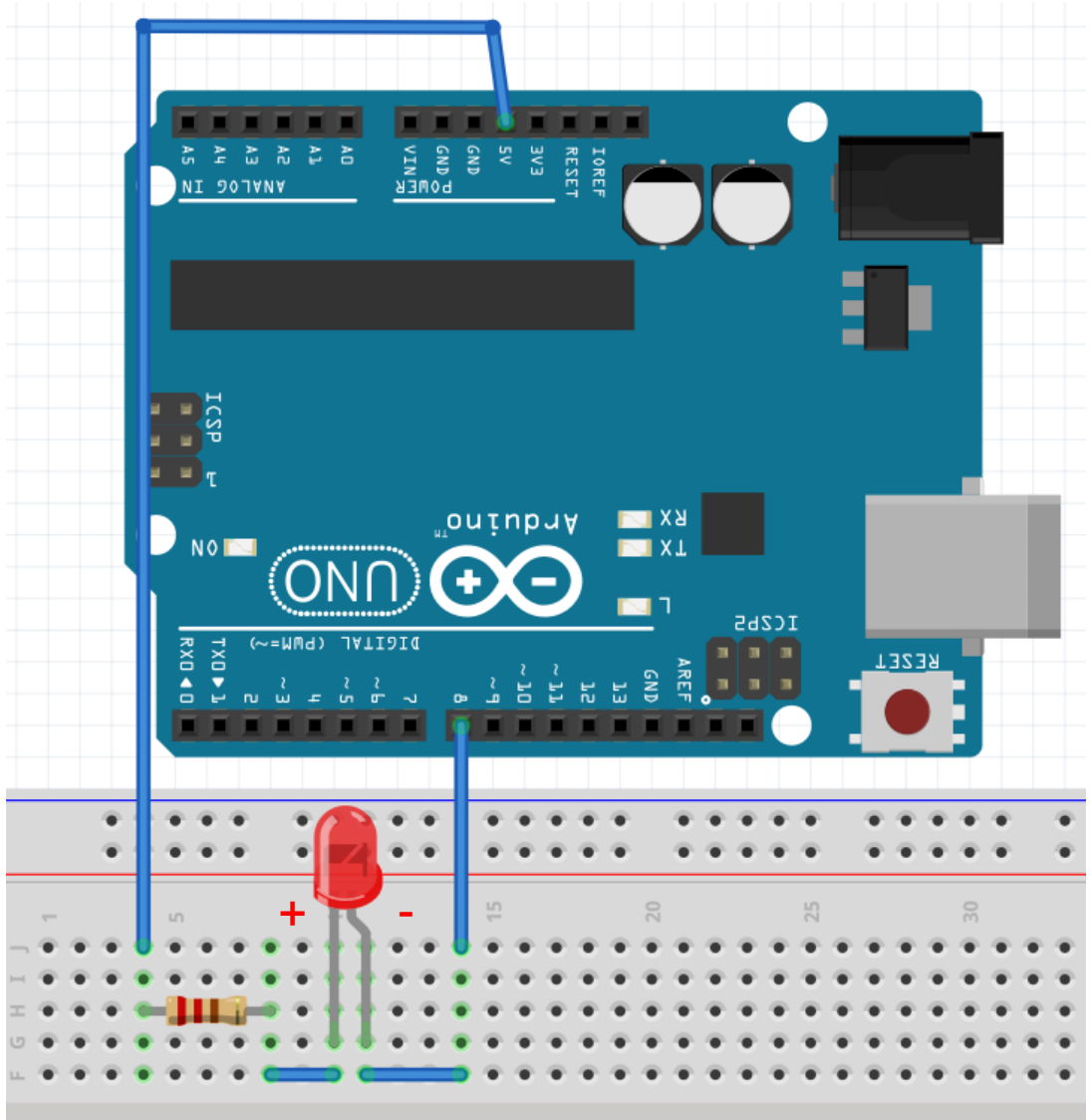
Bit	7	6	5	4	3	2	1	0	
0x09 (0x29)	PIND7	PIND6	PIND5	PIND4	PIND3	PIND2	PIND1	PIND0	PIND
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

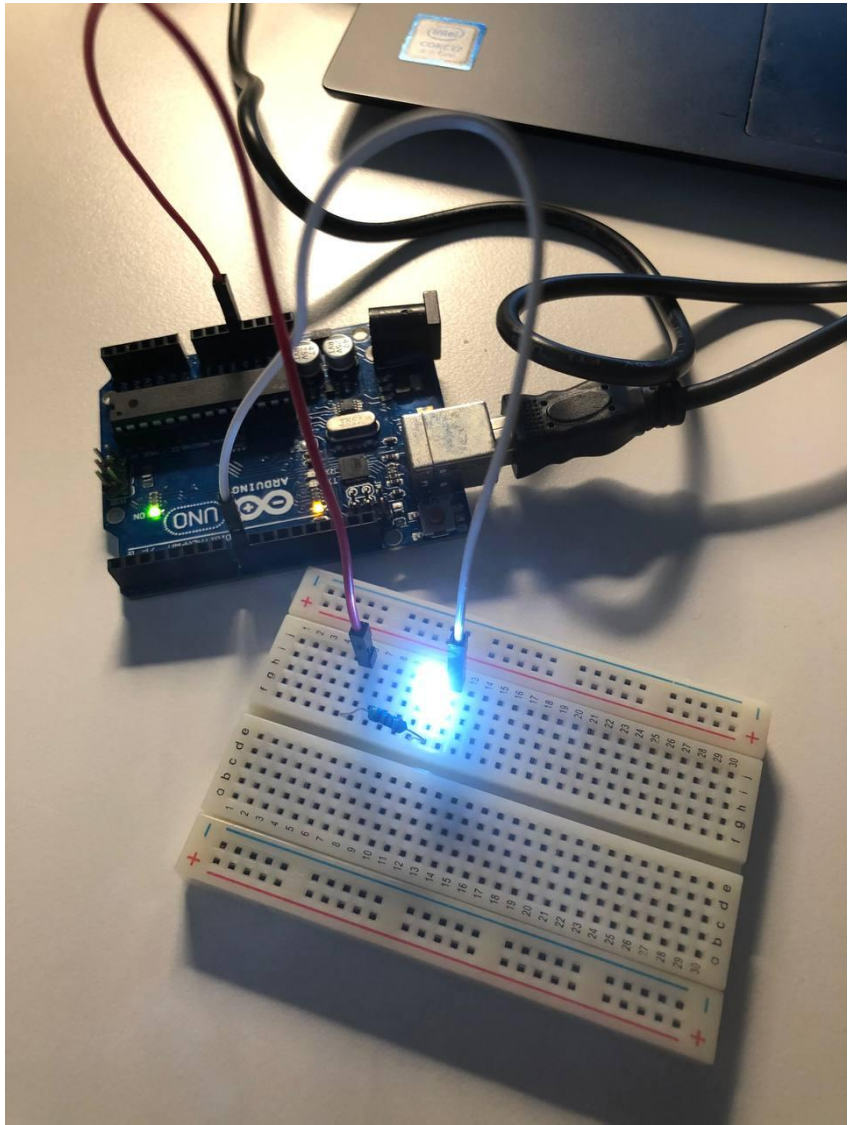
IO 포트 출력 실험

- LED를 이용한 포트 Output 테스트



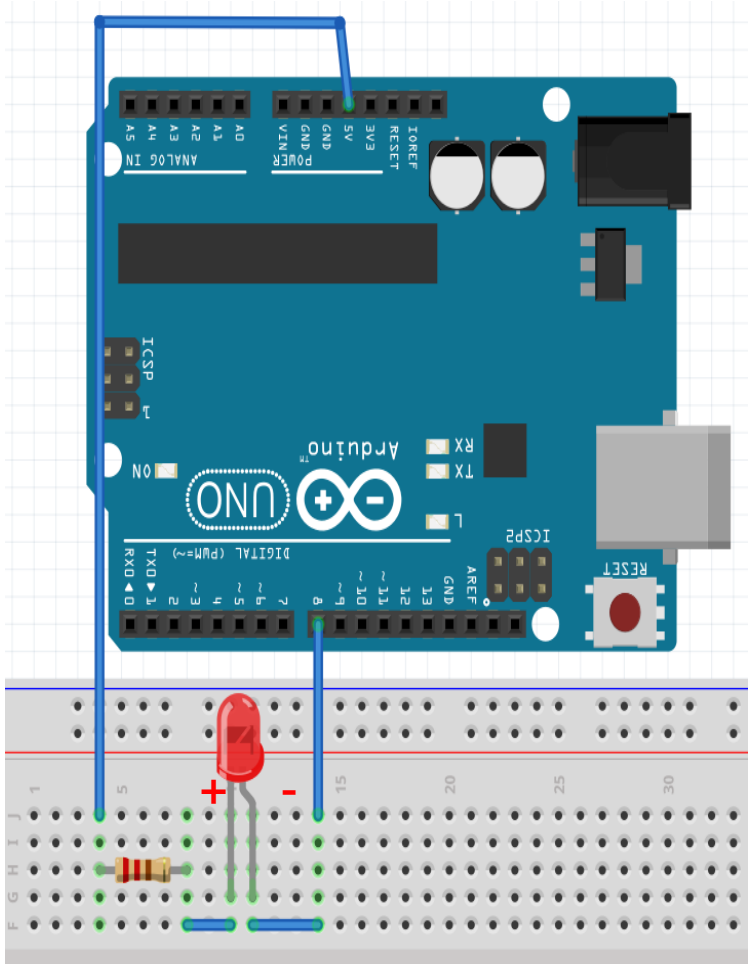
IO 포트 출력 실험

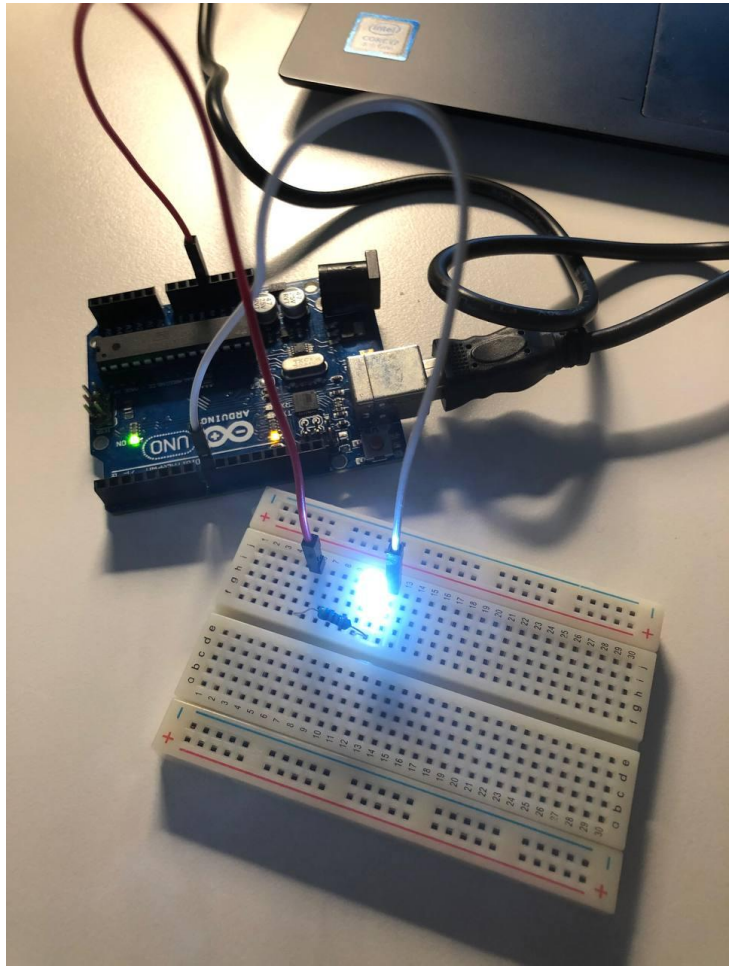

ANALOG IN
A5 A4 A3 A2 A1 A0
POWER
VIN GND GND 5V 3V3 RESET IOREF
ICSP
ON
Arduino
UNO
RX
TX
L
ICSP2
RESET
DIGITAL (PWM=~)
RX0 TX1 2 ~3 4 ~5 ~6 7 8 ~9 ~10 ~11 12 13 GND AREF
+
-



IO 포트 출력 실험

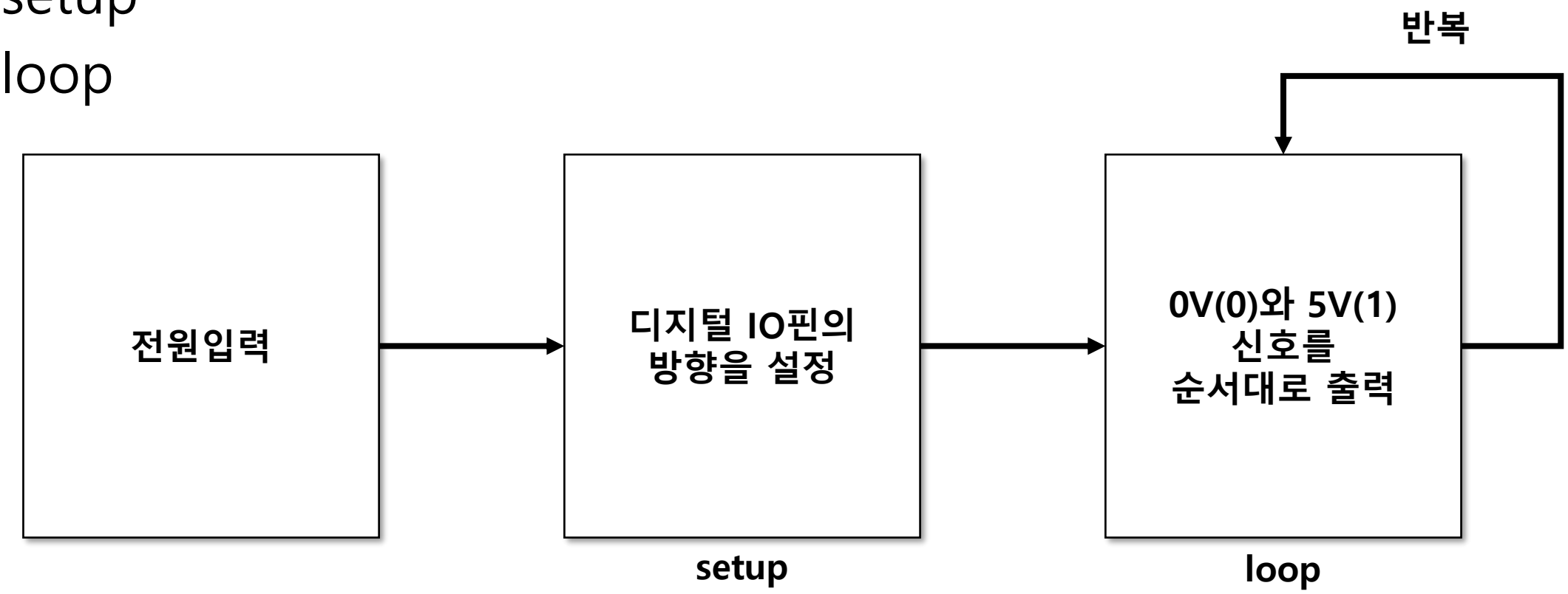
```
void setup()
{
  DDRB = B00000001 ;
  PORTB = B00000000 ;
}

void loop()
{
  PORTB = B00000001 ;
  delay(1000) ;
  PORTB = B00000000 ;
  delay(1000) ;
}
```



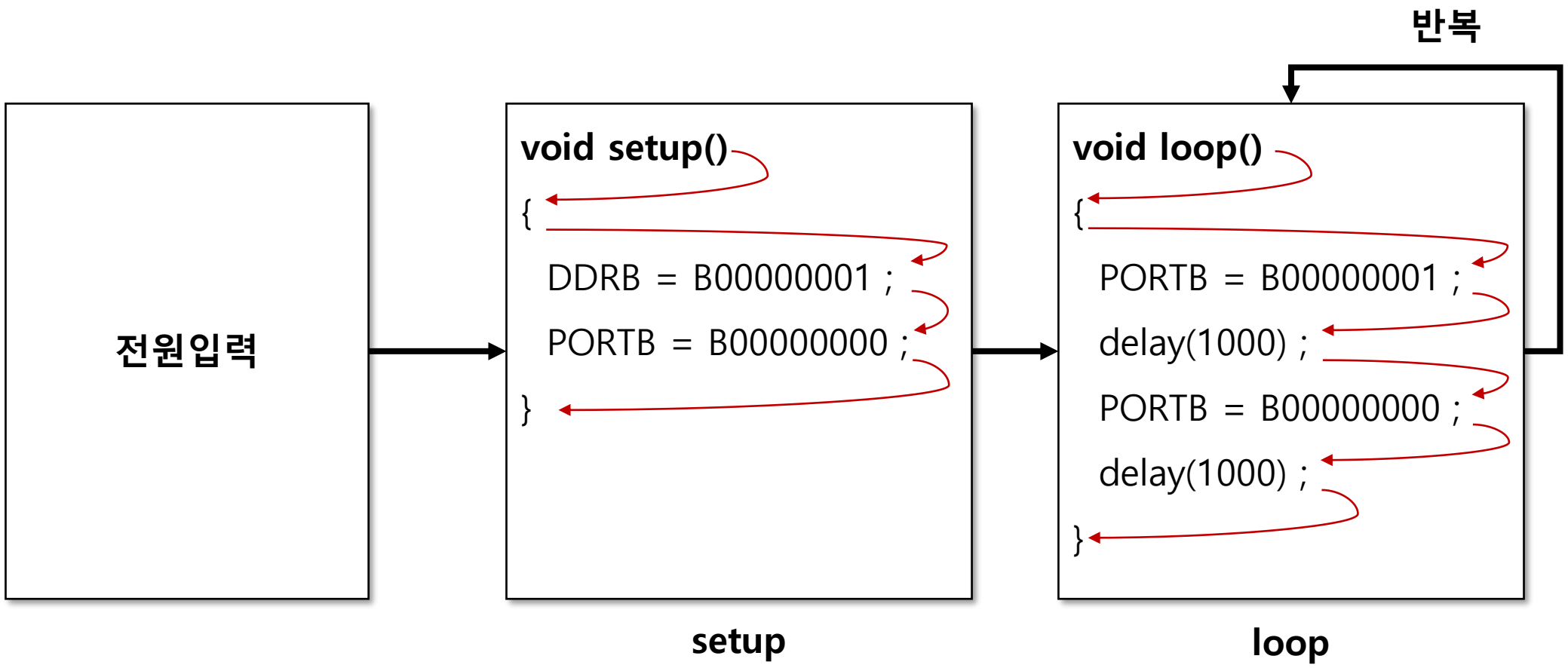


IO 포트 출력 실험

- setup
- loop



IO 포트 출력 실험


반복
전원입력
void setup()
{
DDRB = B00000001 ;
PORTB = B00000000 ;
}
setup
void loop()
{
PORTB = B00000001 ;
delay(1000) ;
PORTB = B00000000 ;
delay(1000) ;
}
loop

IO 포트 출력 실험

DDRB – The Port B Data Direction Register

Bit	7	6	5	4	3	2	1	0	
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

전원입력

```
void setup()
{
  DDRB = B00000001 ;
  PORTB = B00000000 ;
}
```

setup

```
{
  PORTB = B00000001 ;
  delay(1000) ;
  PORTB = B00000000 ;
```

loop

PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

IO 포트 출력 실험

반복

전원입력

```
void setup()
{
  DDRB = B00000001 ;
  PORTB = B00000000 ;
}
```

setup

```
void loop()
{
  PORTB = B00000001 ;
  delay(1000) ;
  PORTB = B00000000 ;
  delay(1000) ;
}
```

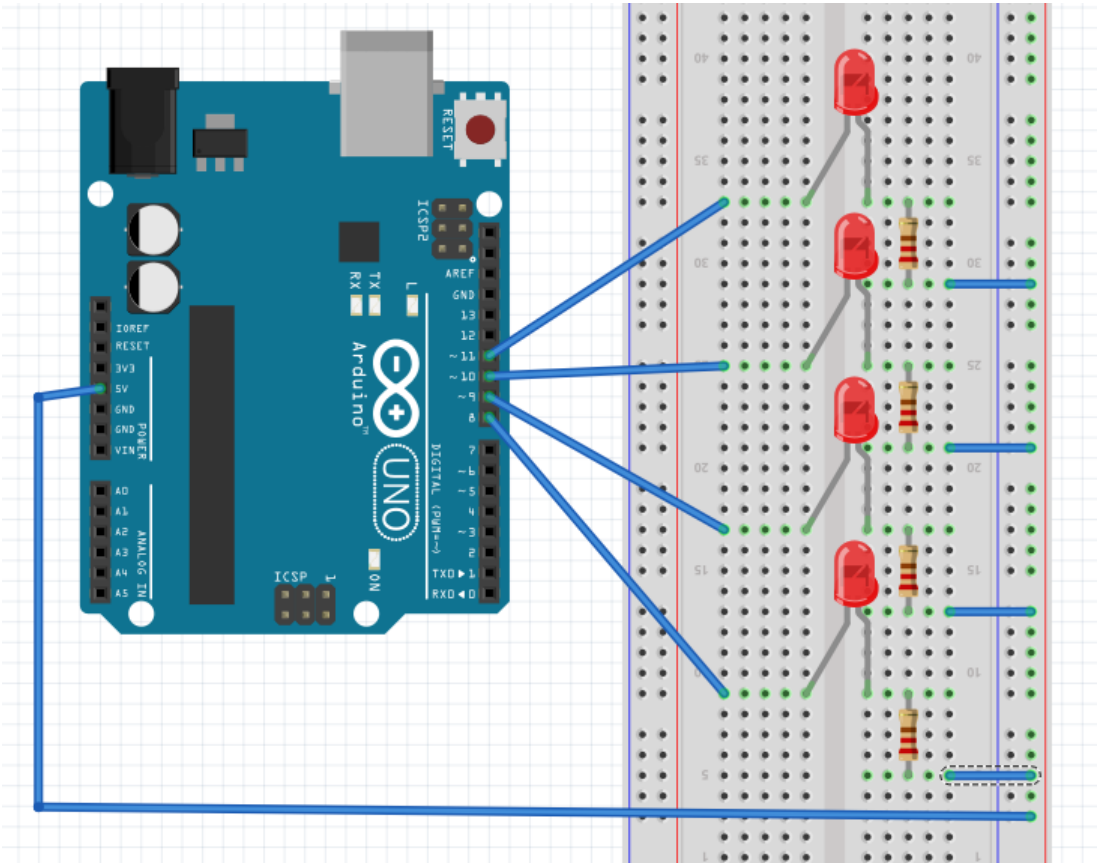
loop

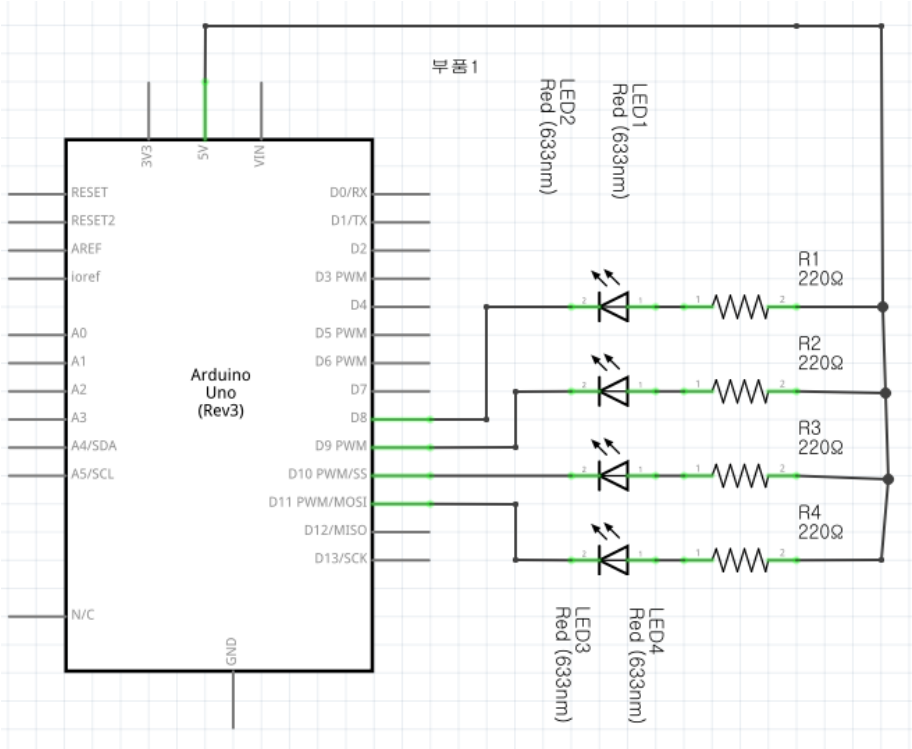
PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0	
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0	PORTB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

IO 포트 출력 실험2

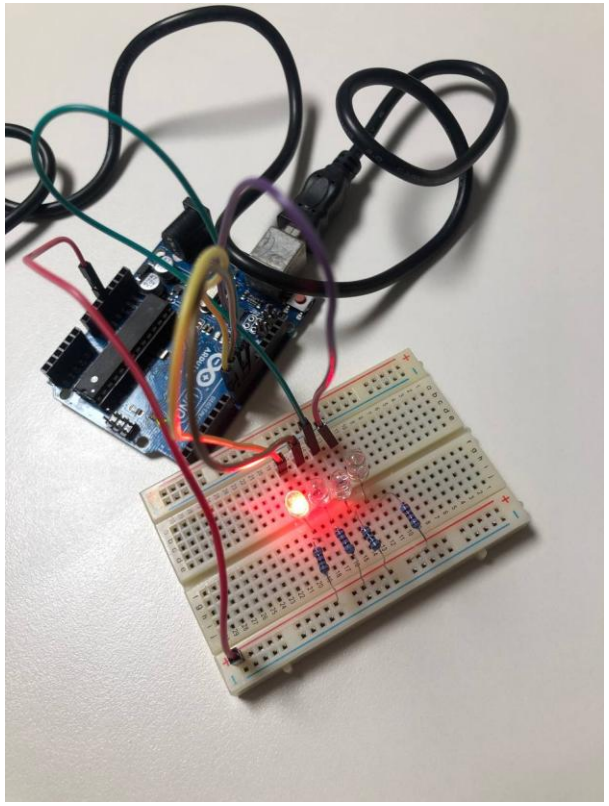
- 4개의 LED를 연결해서 포트를 제어 해보자.



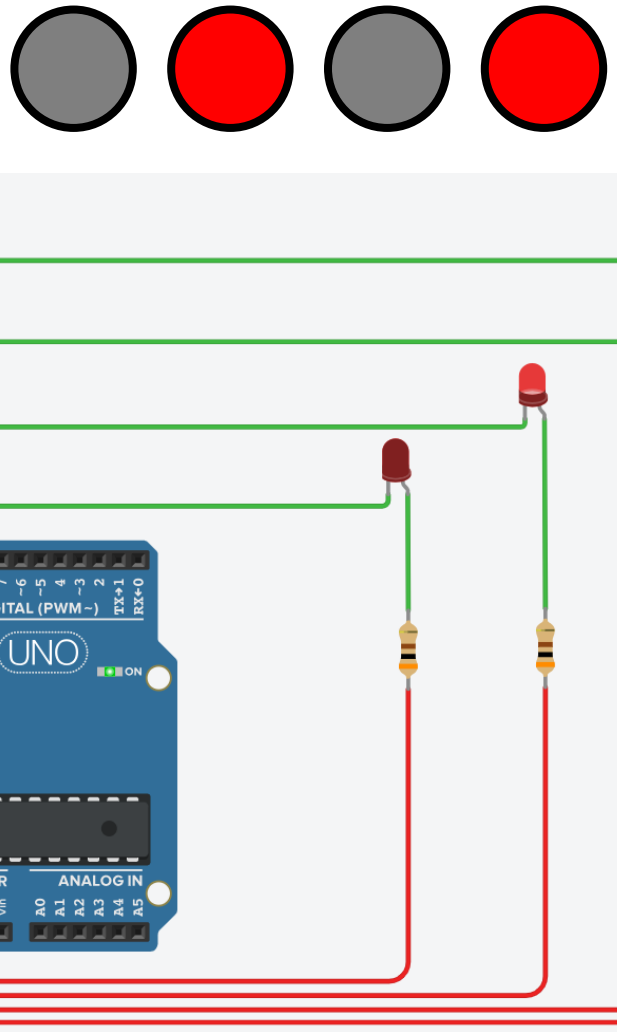


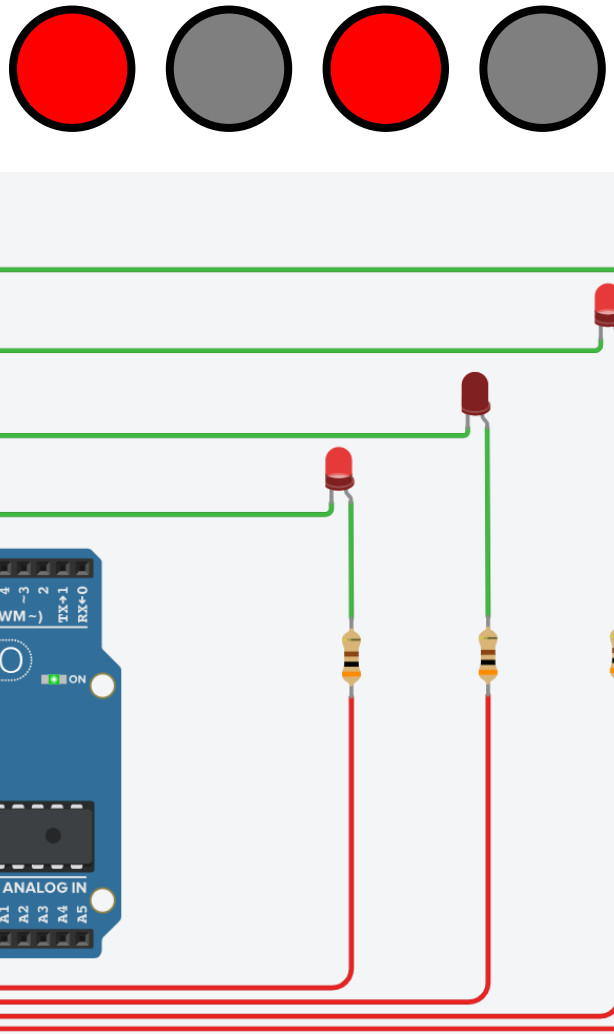
IO 포트 출력 실험2

```
void setup()
{
  DDRB = B00001111 ;
  PORTB = B00000000 ;
}

void loop()
{
  PORTB = B00001111 ;
  delay(1000) ;
  PORTB = B00000000 ;
  delay(1000) ;
}
```



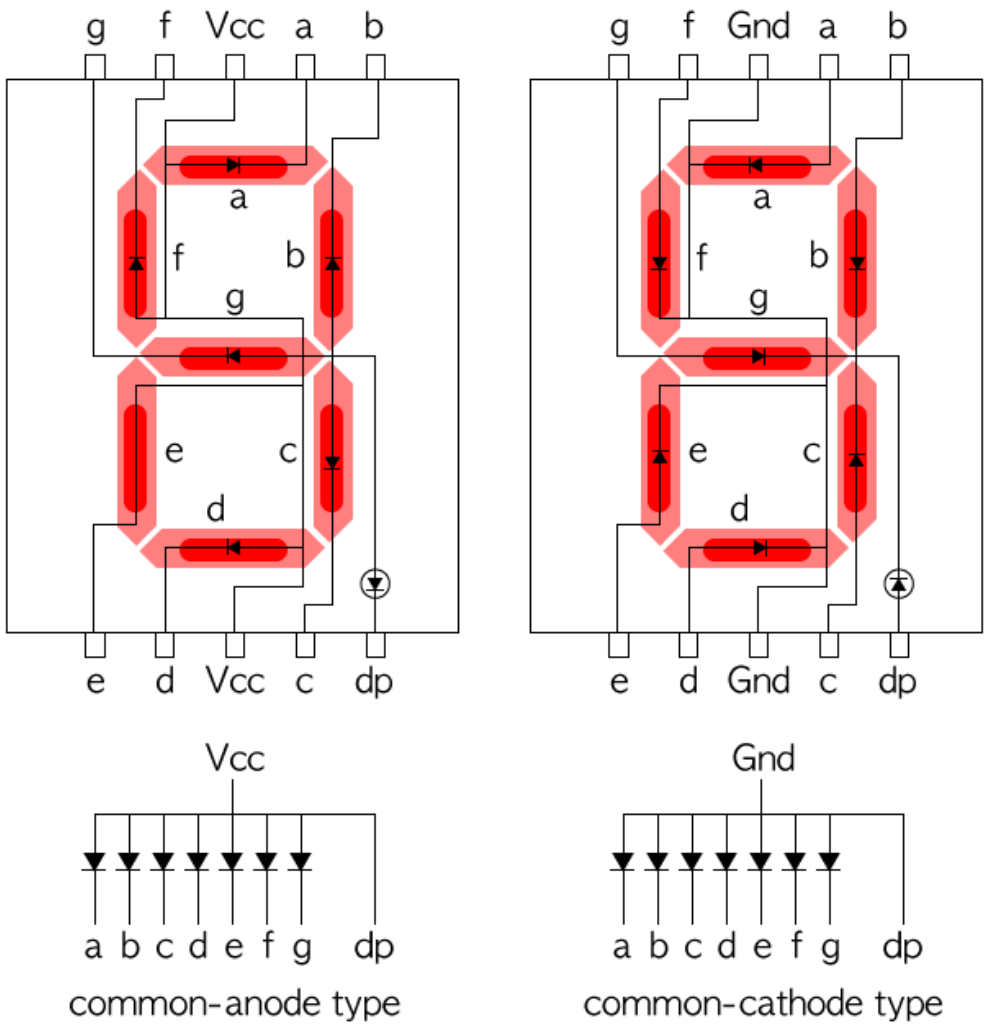
IO 포트 출력 실험3 - QUIZ





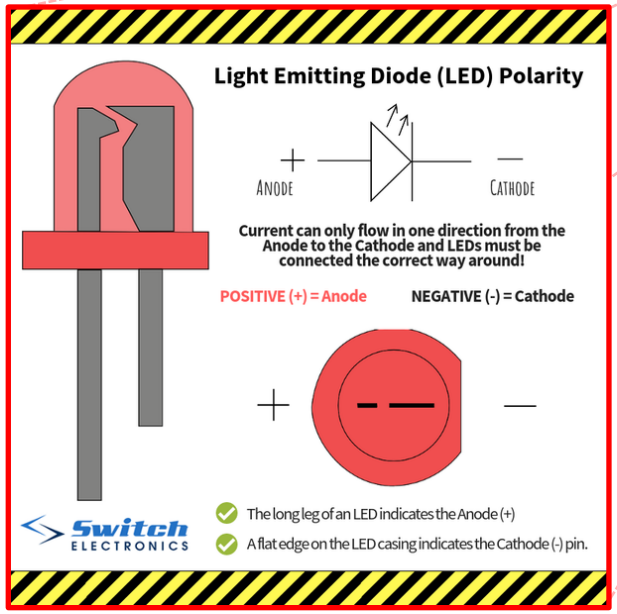
LED를 이용한 포트 제어 실험

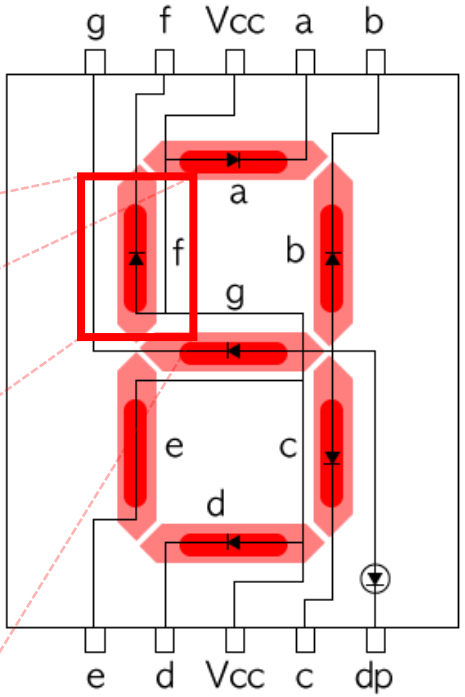
- 7-segment 실험

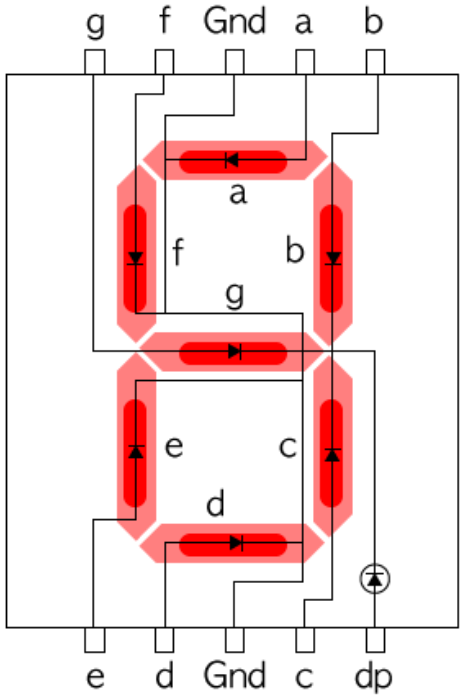


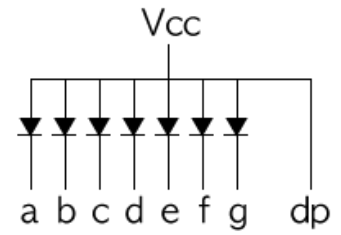
LED를 이용한 포트 제어 실험

- 7-segment 실험

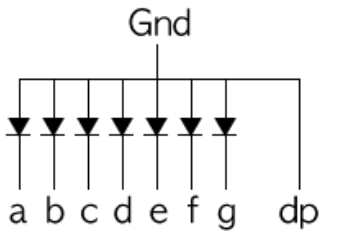








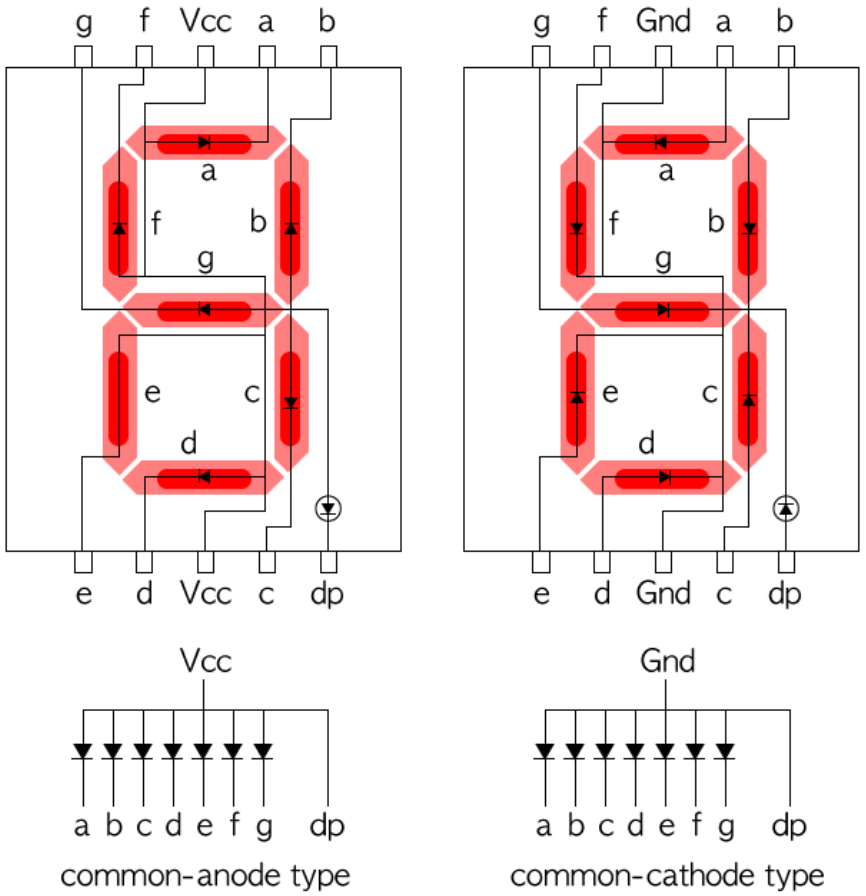
common-anode type



common-cathode type

LED를 이용한 포트 제어 실험

- 7-segment 실험



Common-anode type

0 = a(0) b(0) c(0) d(0) e(0) f(0) g(1) DP(1)

1 = a(1) b(0) c(0) d(1) e(1) f(1) g(1) DP(1)

2 = a(0) b(0) c(1) d(0) e(0) f(1) g(0) DP(1)

3 = a(0) b(0) c(0) d(0) e(1) f(1) g(0) DP(1)

4 = a(1) b(0) c(0) d(1) e(1) f(0) g(0) DP(1)

5 = a(0) b(1) c(0) d(0) e(1) f(0) g(0) DP(1)

6 = a(0) b(1) c(0) d(0) e(0) f(0) g(0) DP(1)

7 = a(0) b(0) c(0) d(1) e(1) f(0) g(1) DP(1)

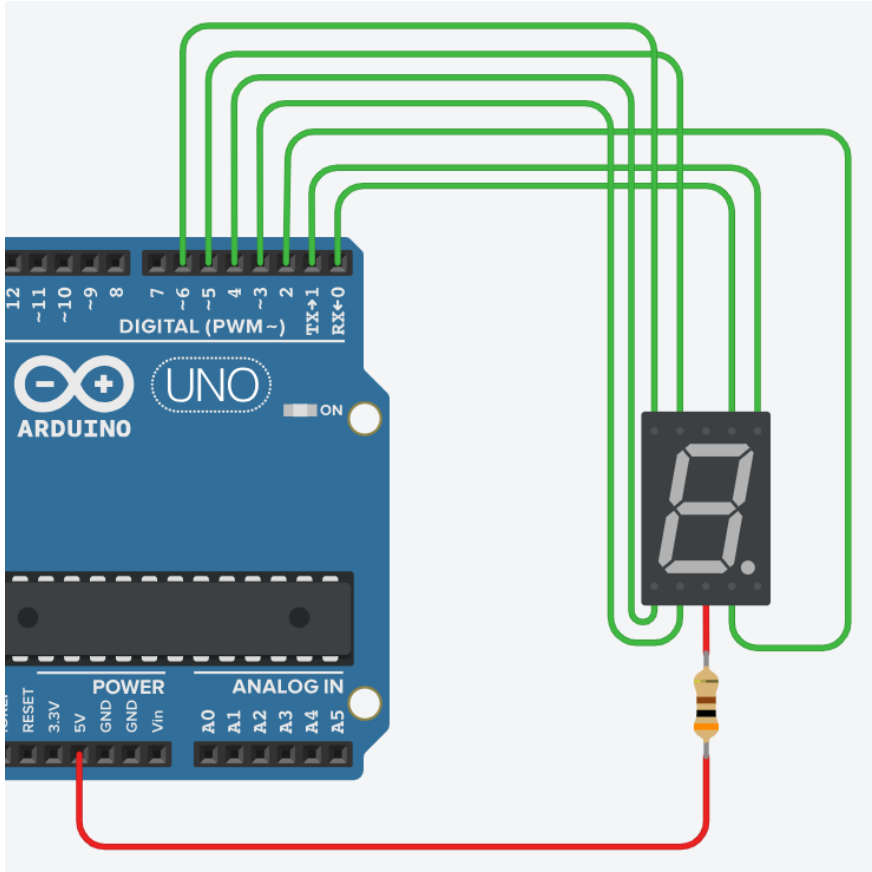
8 = a(0) b(0) c(0) d(0) e(0) f(0) g(0) DP(1)

9 = a(0) b(0) c(0) d(0) e(1) f(0) g(0) DP(1)

. = DP(0)

LED를 이용한 포트 제어 실험

- 7-segment 실험



a	→	PD0
b	→	PD1
c	→	PD2
d	→	PD3
e	→	PD4
f	→	PD5
g	→	PD6
DP	→	PD7

LED를 이용한 포트 제어 실험

- 7-segment 실험

0 = a(0) b(0) c(0) d(0) e(0) f(0) g(1) DP(1)

1 = a(1) b(0) c(0) d(1) e(1) f(1) g(1) DP(1)

2 = a(0) b(0) c(1) d(0) e(0) f(1) g(0) DP(1)

3 = a(0) b(0) c(0) d(0) e(1) f(1) g(0) DP(1)

4 = a(1) b(0) c(0) d(1) e(1) f(0) g(0) DP(1)

5 = a(0) b(1) c(0) d(0) e(1) f(0) g(0) DP(1)

6 = a(0) b(1) c(0) d(0) e(0) f(0) g(0) DP(1)

7 = a(0) b(0) c(0) d(1) e(1) f(0) g(1) DP(1)

8 = a(0) b(0) c(0) d(0) e(0) f(0) g(0) DP(1)

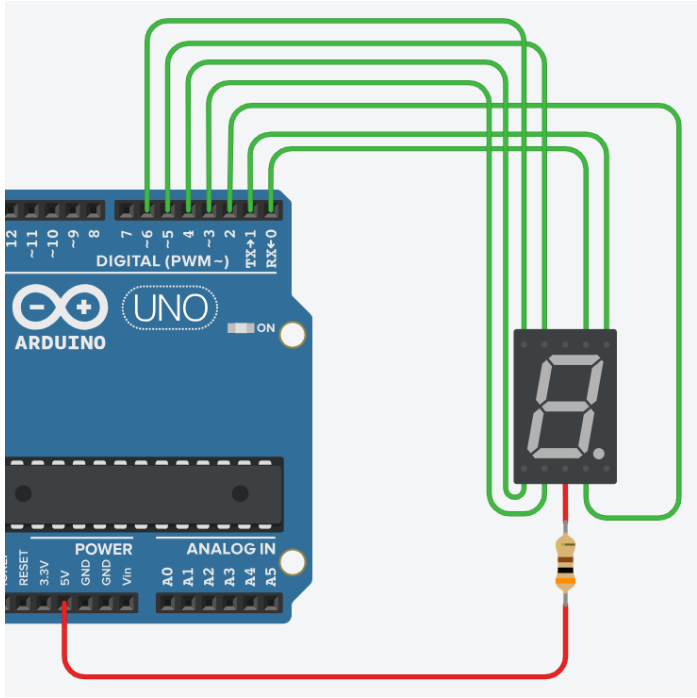
9 = a(0) b(0) c(0) d(0) e(1) f(0) g(0) DP(1)

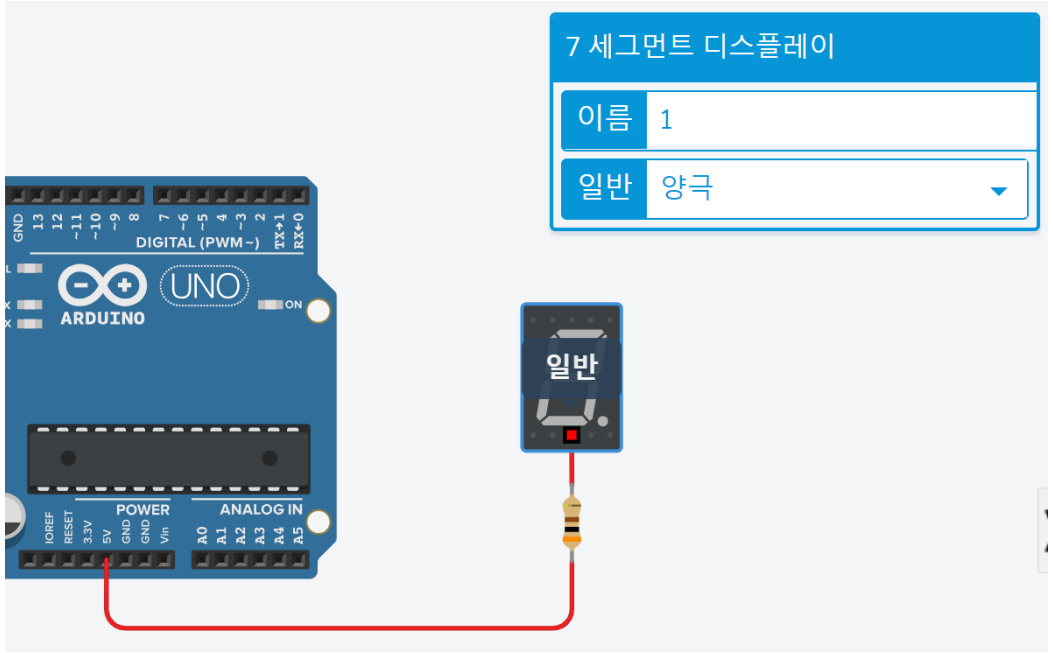
. = DP(0)

		a	b	c	d	e	f	g	DP
		PD0	PD1	PD2	PD3	PD4	PD5	PD6	PD7
0	→	0	0	0	0	0	0	1	1
1	→	1	0	0	1	1	1	1	1
2	→	0	0	1	0	0	1	0	1
3	→	0	0	0	0	1	1	0	1
4	→	1	0	0	1	1	0	0	1
5	→	0	1	0	0	1	0	0	1
6	→	0	1	0	0	0	0	0	1
7	→	0	0	0	1	1	0	1	1
0	→	0	0	0	0	0	0	0	1
9	→	0	0	0	0	1	0	0	1

LED를 이용한 포트 제어 실험

- 7-segment 실험



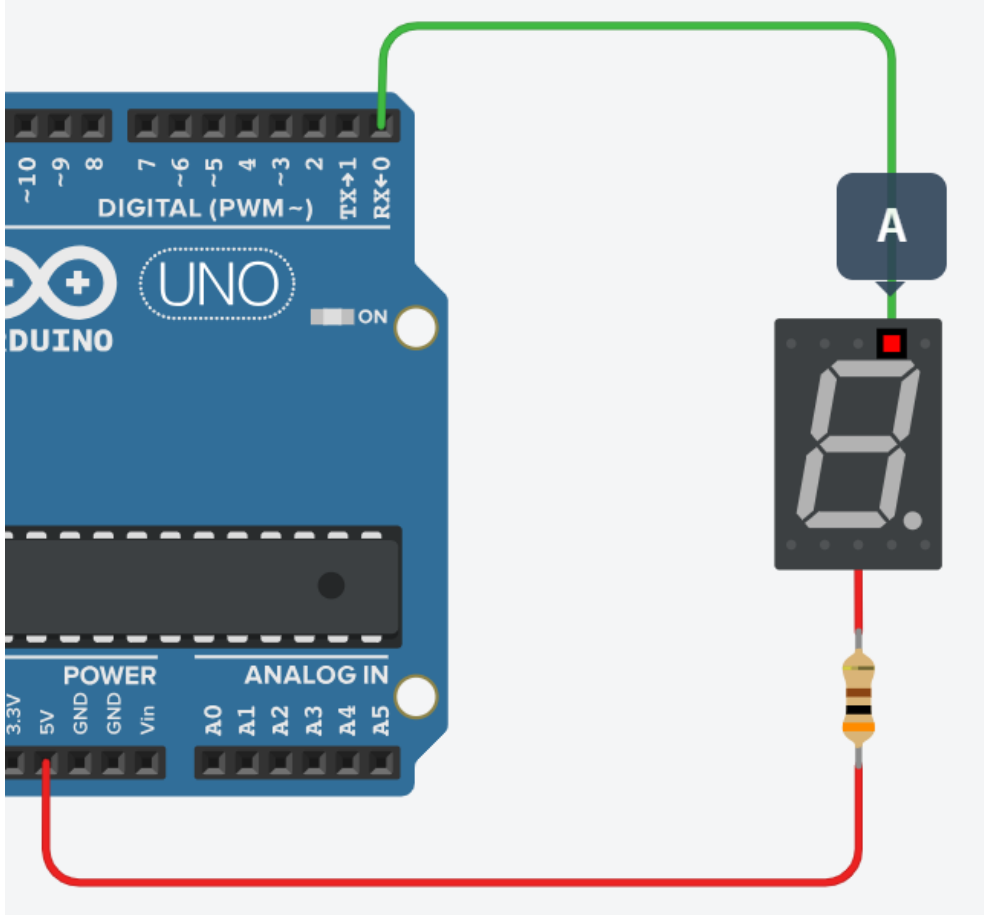


LED를 이용한 포트 제어 실험

- 7-segment 실험

```
void setup()
{
  DDRD = B00000001 ;
}

void loop()
{
 PORTD = B00000000 ;
}
```

```
void setup()
{
  DDRD = B00000001 ;
}

void loop()
{
 PORTD = B00000001 ;
}
```

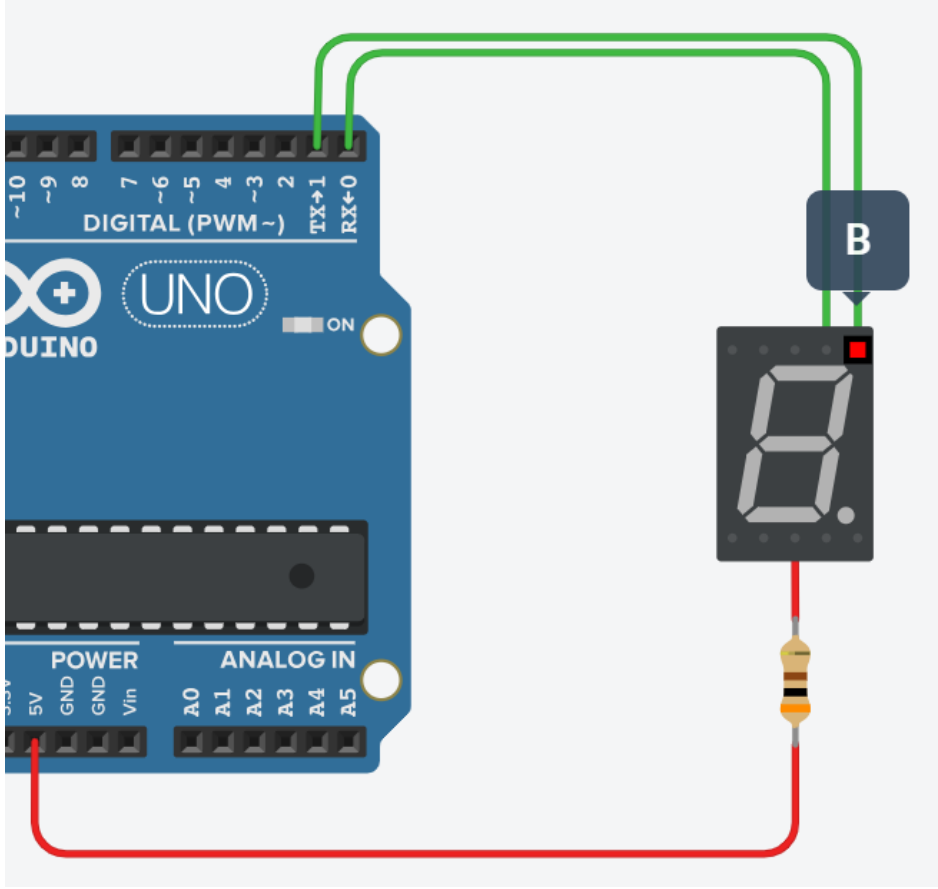


LED를 이용한 포트 제어 실험

- 7-segment 실험

```
void setup()
{
  DDRD = B00000011 ;
}

void loop()
{
 PORTD = B00000000 ;
}
```

```
void setup()
{
  DDRD = B00000011 ;
}

void loop()
{
 PORTD = B00000011 ;
}
```

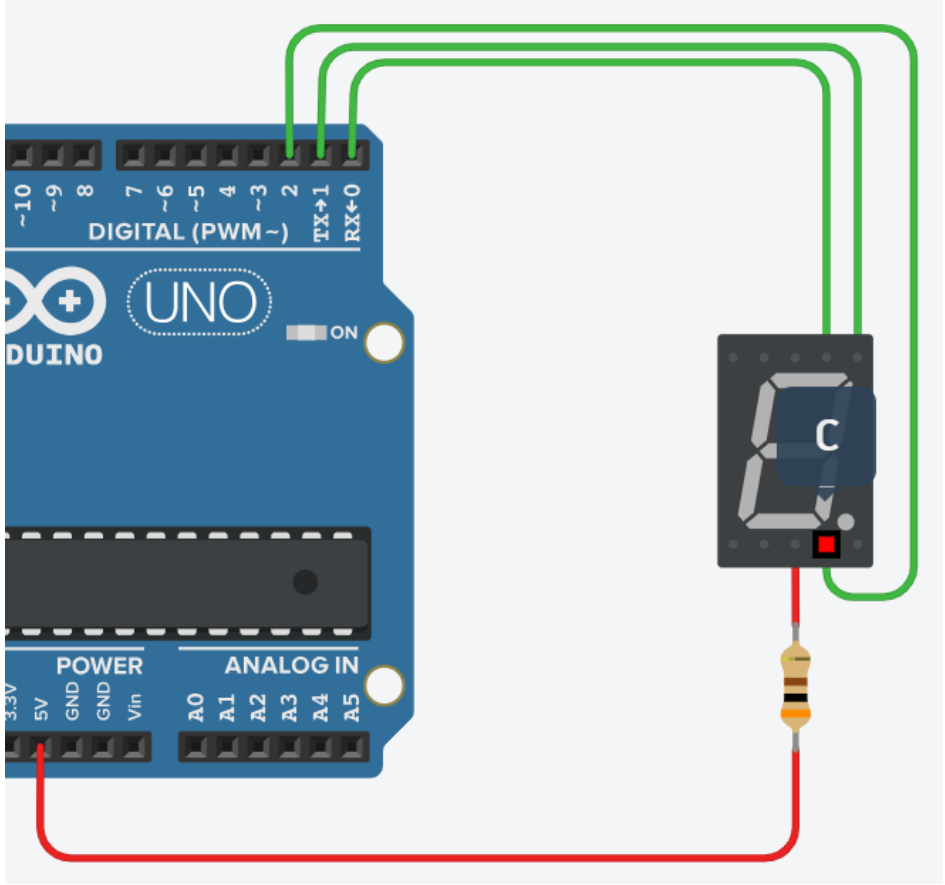


LED를 이용한 포트 제어 실험

- 7-segment 실험

```
void setup()
{
  DDRD = B00000111 ;
}

void loop()
{
 PORTD = B00000000 ;
}
```

```
void setup()
{
  DDRD = B00000111 ;
}

void loop()
{
 PORTD = B00000111 ;
}
```

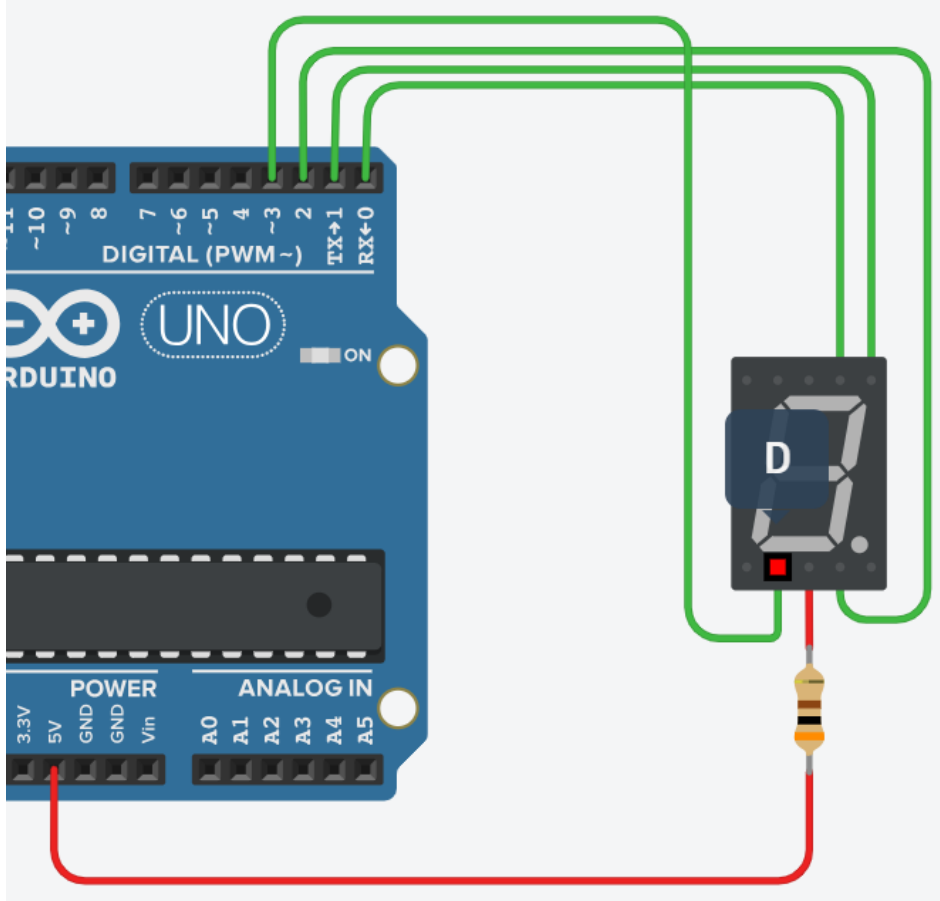


LED를 이용한 포트 제어 실험

- 7-segment 실험

```
void setup()
{
  DDRD = B00001111 ;
}

void loop()
{
 PORTD = B00000000 ;
}
```

```
void setup()
{
  DDRD = B00001111 ;
}

void loop()
{
 PORTD = B00001111 ;
}
```

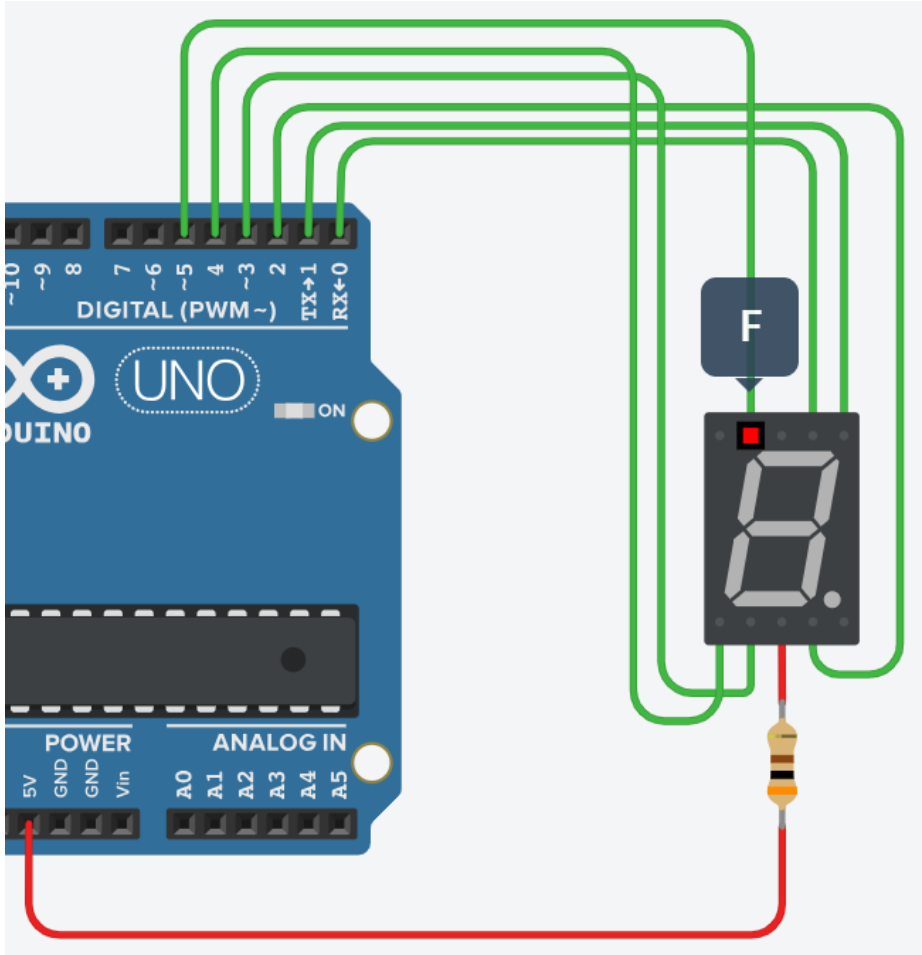


LED를 이용한 포트 제어 실험

- 7-segment 실험

```
void setup()
{
  DDRD = B00111111 ;
}

void loop()
{
 PORTD = B00000000 ;
}
```

```
void setup()
{
  DDRD = B00111111 ;
}

void loop()
{
 PORTD = B00111111 ;
}
```

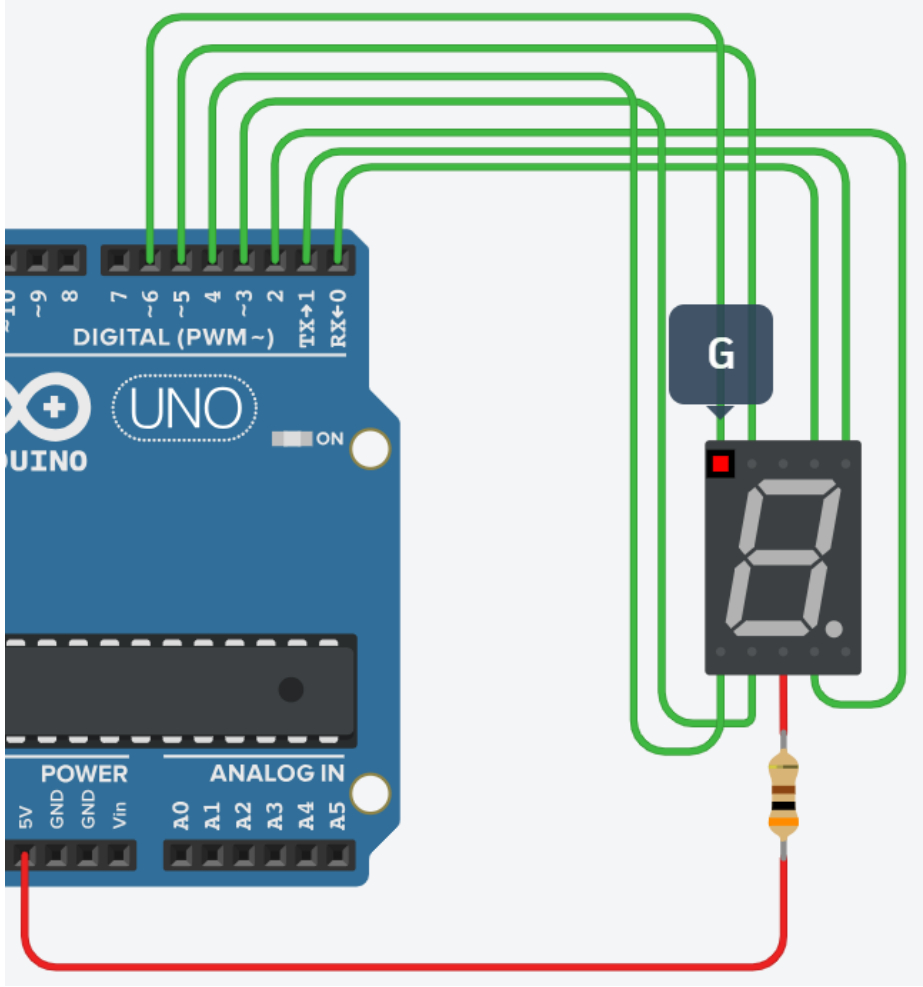


LED를 이용한 포트 제어 실험

- 7-segment 실험

```
void setup()
{
  DDRD = B01111111 ;
}

void loop()
{
 PORTD = B00000000 ;
}
```

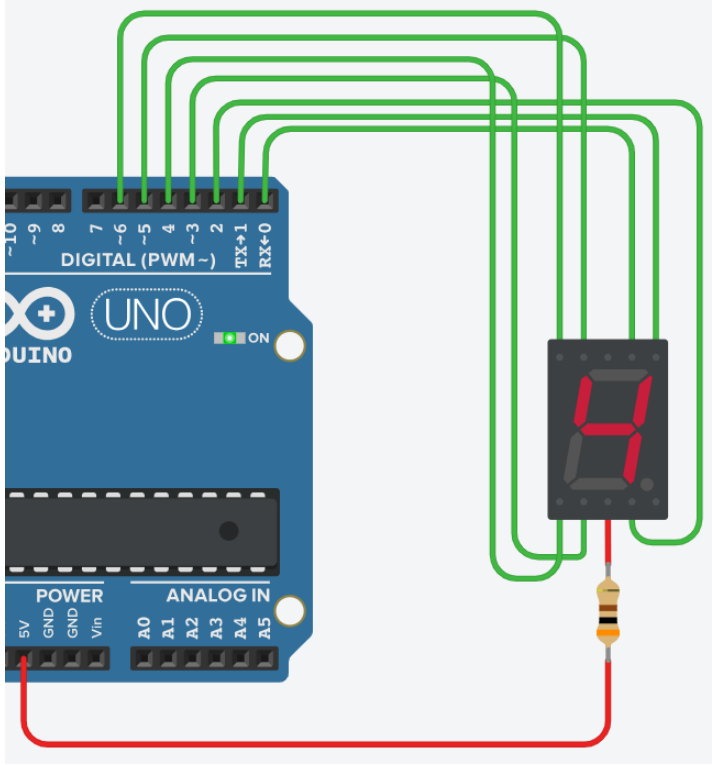
```
void setup()
{
  DDRD = B01111111 ;
}

void loop()
{
 PORTD = B01111111 ;
}
```



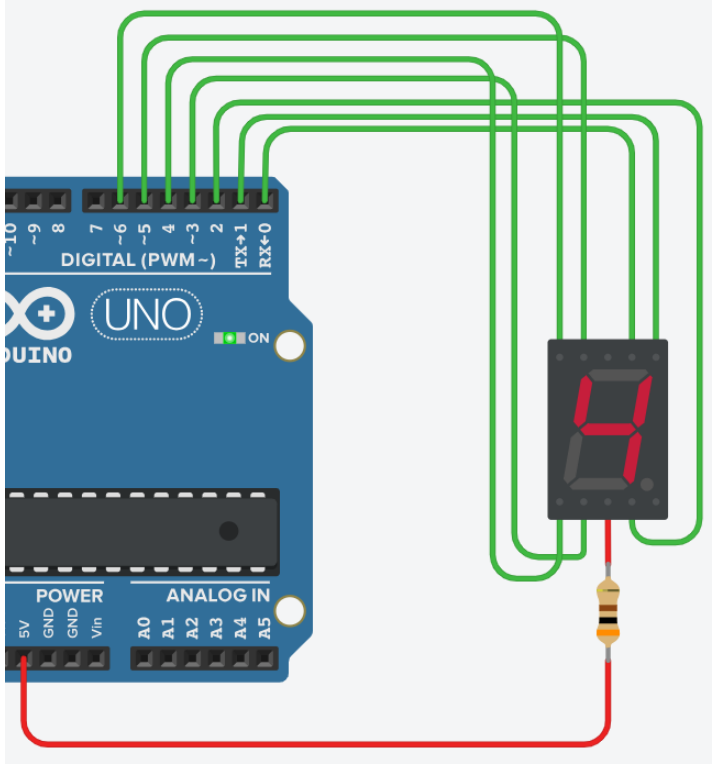
LED를 이용한 포트 제어 실험

- 7-segment 실험 QUIZ – 숫자 **4**를 표시 하시오.

```
void setup()
{
  DDRD = B[_______________] ;
}

void loop()
{
 PORTD = B[_______________]  ;
}
```



LED를 이용한 포트 제어 실험

- 7-segment 실험 QUIZ – 숫자 **4**를 표시 하시오.

```
void setup()
{
  DDRD = 0x[_____________] ;
}

Void loop()
{
 PORTD = 0x[_____________]  ;
}
```